User Test Geometry

1. Executive Summary

Total Floor Area	120.00 m²
Stories / Height	2 Stories / 6.0 m
RCC Concrete Volume	3.24 m³
Reinforcement Steel	505 kg
Clay Modular Bricks	17,512 Nos
Masonry & Plaster Cement	152 Bags (50kg)
Masonry & Plaster Sand	36.46 Tonnes
Plaster Surface Area	608.9 m²
Total Estimated Cost	INR 1,043,918.82

2. Design Basis

Design Codes & Standards:
- IS 456:2000 (Code of Practice for Plain and Reinforced Concrete)

Material Properties:
- Concrete: M25 (Density: 25 kN/m3, E = 5000sqrt(fck))
- Steel: Fe415 (Density: 78.5 kN/m3, E = 2x10^5 MPa)

Loading & Analysis:
- Design Load: User Input (Factored)
- Load Path: Tributary Area Method (Corner: 25%, Edge: 50%, Interior: 100% of bay)
- Beam Analysis: Simplified Simply Supported/Continuous assumptions.
(M = wL^2/8 or wL^2/10, Deflection checked against L/250)

Column Design:
- Short Column Assumption (Le/r < 12)
- Sizing Criteria: Axial Stress Check (Pu/Ag <= 0.4 fck)
Note: Steel capacity (0.67 fy Asc) is reserved as safety margin.

Foundation Design:
- Isolated Square Footings
- Area Required = Load * 1.1 / SBC (1.1 factor for self-weight)
- Depth Check: Punching Shear at d/2 distance (Limit: 0.25sqrt(fck))

Estimation Basis:
- Concrete Cost: INR 5000/m3
- Steel Cost: INR 60/kg
- Rebar Ratios: Columns 150kg/m3, Beams 120kg/m3, Footings 80kg/m3

3. Column Schedule & Quantities (IS 456 Cl. 26.5.3)

ID	Level	Loc (x,y)	Load (kN)	Size (mm)	Conc (m3)	Steel (kg)	Status
C0_L0	L0	(0.00, 0.00)	0.0	300x300	0.27	40.5	PASS
C4_L0	L0	(0.00, 4.00)	0.0	300x300	0.27	40.5	PASS
C8_L0	L0	(0.00, 8.00)	0.0	300x300	0.27	40.5	PASS
C1_L0	L0	(3.30, 0.00)	0.0	300x300	0.27	40.5	PASS
C5_L0	L0	(3.30, 4.00)	0.0	300x300	0.27	40.5	PASS

ID	Level	Loc (x,y)	Load (kN)	Size (mm)	Conc (m3)	Steel (kg)	Status
C9_L0	L0	(3.30, 8.00)	0.0	300x300	0.27	40.5	PASS
C2_L0	L0	(6.60, 0.00)	0.0	300x300	0.27	40.5	PASS
C6_L0	L0	(6.60, 4.00)	0.0	300x300	0.27	40.5	PASS
C10_L0	L0	(6.60, 8.00)	0.0	300x300	0.27	40.5	PASS
C3_L0	L0	(9.90, 0.00)	0.0	300x300	0.27	40.5	PASS
C7_L0	L0	(9.90, 4.00)	0.0	300x300	0.27	40.5	PASS
C11_L0	L0	(9.90, 8.00)	0.0	300x300	0.27	40.5	PASS

4. Reinforcement Schedule (IS 456 Cl. 26.5)

Col ID	Main Bars	Links (Stirrups)
N/A	Run Detailing	First

5. Foundation Schedule & Quantities (IS 456 Cl. 34)

Col ID	Load (kN)	Size (m)	Depth (mm)	Conc (m3)	Steel (kg)
N/A	-	-	-	-	-

6. Beam Schedule & Quantities (IS 456 Cl. 26.5.1)

ID	Size (mm)	Span (m)	Conc (m3)	Steel (kg)	Top Steel	Bot Steel	Stirrups
N/A	-	-	-	-	-	-	-

7. Slab Schedule & Quantities

ID	Thk (mm)	Area (m2)	Conc (m3)	Steel (kg)	Main Mesh	Dist Mesh
N/A	-	-	-	-	-	-

8. Brickwork Schedule & Material Takeoff (IS 1077, IS 2250)

Modular clay bricks 200×200×100 mm with mortar (190×190×90 mm solid core, 250 bricks/m³ per IS 1077). Mortar 1:4 mix proportion with 1.33 dry volume factor per IS 2250. W:C ratio = 0.50.

Wall ID	Span / Cols	Len (m)	Ht (m)	Thk	Gross (m³)	Bricks (Nos)	Brick Vol	Wet Mor	Dry Mor	Cem Bags	Cem (kg)	Sand (T)	Water (L)
W1	Bay	10.00	2.48	230mm	11.39	2,848	9.25	2.132	2.835	17	816.6	3.63	408
W2	Bay	12.00	2.48	230mm	13.66	3,416	11.10	2.563	3.409	20	981.9	4.36	491
W3	Bay	10.00	2.48	230mm	11.39	2,848	9.25	2.132	2.835	17	816.6	3.63	408
W4	Bay	12.00	2.48	230mm	13.66	3,416	11.10	2.563	3.409	20	981.9	4.36	491
W5	Bay	2.50	2.48	230mm	2.85	712	2.31	0.533	0.709	5	204.1	0.91	102
W6	Bay	5.00	2.48	230mm	5.69	1,424	4.63	1.066	1.418	9	408.3	1.81	204
W7	Bay	2.50	2.48	230mm	2.85	712	2.31	0.533	0.709	5	204.1	0.91	102
W8	Bay	7.50	2.48	230mm	8.54	2,136	6.94	1.599	2.127	13	612.4	2.72	306
TOTAL	All Walls	-	-	-	70.02	17,512	56.90	13.12	17.45	101	5,026	22.34	2,513

9. Plastering Schedule & Quantities (IS 1661, IS 2402, IS 1200)

External wall plastering: 15mm thick in 1:4 cement:sand mix (IS 2402). Internal wall plastering: 12mm thick in 1:6 mix (IS 1661). Ceiling soffit plastering: 6mm thick in 1:3 mix per IS 1661 Cl. 12.3. Dry volume factor = 1.33.

Surface ID	Location & Mix Spec	Area (m²)	Thk (mm)	Mix	Wet Vol (m³)	Dry Vol (m³)	Cem (Bags)	Cem (kg)	Sand (T)	Water (L)
W1_EXT	Ext Render (15mm 1:4)	49.5	15	1:4	0.742	0.988	6	284.4	1.26	142
W1_INT	Int Wall (12mm 1:6)	49.5	12	1:6	0.594	0.790	4	162.5	1.08	81
W2_EXT	Ext Render (15mm 1:4)	59.4	15	1:4	0.891	1.185	7	341.3	1.52	171
W2_INT	Int Wall (12mm 1:6)	59.4	12	1:6	0.713	0.948	4	195.0	1.30	98
W3_EXT	Ext Render (15mm 1:4)	49.5	15	1:4	0.742	0.988	6	284.4	1.26	142
W3_INT	Int Wall (12mm 1:6)	49.5	12	1:6	0.594	0.790	4	162.5	1.08	81
W4_EXT	Ext Render (15mm 1:4)	59.4	15	1:4	0.891	1.185	7	341.3	1.52	171
W4_INT	Int Wall (12mm 1:6)	59.4	12	1:6	0.713	0.948	4	195.0	1.30	98
W5	Int Part (2-Face 12mm 1:6)	24.8	12	1:6	0.297	0.395	2	81.3	0.54	41
W6	Int Part (2-Face 12mm 1:6)	49.5	12	1:6	0.594	0.790	4	162.5	1.08	81
W7	Int Part (2-Face 12mm 1:6)	24.8	12	1:6	0.297	0.395	2	81.3	0.54	41
W8	Int Part (2-Face 12mm 1:6)	74.2	12	1:6	0.891	1.185	5	243.8	1.63	122
Ceiling Soffit	Slab Bottom (6mm 1:3)	240.0	6	1:3	1.440	1.915	14	689.5	2.30	345
TOTAL	All Plaster Surfaces	848.9	-	-	9.400	12.501	65	3,225	16.42	1,612

10. Painting & Surface Finishes Schedule (NBC 2016 SP7)

Surface preparation & coatings per NBC 2016 SP7: Water-thinnable primer (1 coat @ 8 m²/L), polymer wall putty (2 coats @ 1.5 kg/m²), interior acrylic emulsion (2 coats @ 12 m²/L), exterior weather-guard acrylic emulsion (2 coats @ 10 m²/L).

Surface ID	Location & Specification	Finish Area (m²)	Primer 1-Ct (L)	Putty 2-Ct (kg)	Int Emulsion 2-Ct (L)	Ext Weather 2-Ct (L)
W1_EXT	External Weather Guard	49.5	6.2	—	—	9.9
W1_INT	Internal Acrylic Emulsion	49.5	6.2	74.2	8.2	—
W2_EXT	External Weather Guard	59.4	7.4	—	—	11.9
W2_INT	Internal Acrylic Emulsion	59.4	7.4	89.1	9.9	—
W3_EXT	External Weather Guard	49.5	6.2	—	—	9.9
W3_INT	Internal Acrylic Emulsion	49.5	6.2	74.2	8.2	—
W4_EXT	External Weather Guard	59.4	7.4	—	—	11.9
W4_INT	Internal Acrylic Emulsion	59.4	7.4	89.1	9.9	—
W5	Internal Partition Emulsion	24.8	3.1	37.1	4.1	—
W6	Internal Partition Emulsion	49.5	6.2	74.2	8.2	—
W7	Internal Partition Emulsion	24.8	3.1	37.1	4.1	—
W8	Internal Partition Emulsion	74.2	9.3	111.4	12.4	—
Ceiling Soffit	Slab Underside (White Emulsion)	240.0	30.0	360.0	40.0	—
TOTAL	All Surfaces (Walls + Ceilings)	848.9	106.1	946.6	105.2	43.6

11. Consolidated Masonry & Finishes BOQ & Cost (IS 1905, CPWD DSR)

11.1 Comprehensive Bill of Materials

Item / Material Description	Specification / Mix Proportion	Building Total	Unit	IS Standard
Total Wall Gross Volume	L × t × Hv (clear story height)	70.02	m³	IS 1905:1987
Clay Modular Building Bricks	Standard modular 20×20×10 cm	17,512	Nos	IS 1077:1992
Net Solid Brickwork Volume	Actual brick body volume (excl. mortar)	56.90	m³	IS 1077:1992
Wet Brickwork Mortar Volume	Gross Wall Volume – Solid Brick Vol	13.12	m³	IS 2250:1981
Dry Brickwork Mortar Volume	Wet Volume × 1.33 dry factor	17.45	m³	IS 2250:1981
Brickwork Cement	1:4 Mix Proportion (OPC / PPC)	101 (5,026 kg)	Bags (50kg)	IS 269:2015
Brickwork Sand	Coarse sand, dry density 1600 kg/m³	22.34	Tonnes	IS 383 Zone II
Brickwork Water	Water:Cement ratio = 0.50	2,513	Litres	IS 456 Cl 5.4
Wall Plastering Surface Area	Internal (12mm 1:6) + External (15mm 1:4)	608.9	m²	IS 1200 Pt 12
Ceiling Soffit Plaster Area	Soffit bottom of slabs (6mm 1:3 mix)	240.0	m²	IS 1661:1972
Total Plaster Surface Area	Combined wall + ceiling surface	848.9	m²	IS 1200 Pt 12
Plastering Cement	Combined wall & ceiling plaster	65 (3,225 kg)	Bags (50kg)	IS 1661 / IS 2402
Plastering Sand	Clean plaster sand (Zone III/IV)	16.42	Tonnes	IS 383:2016
Plastering Water	Water:Cement ratio = 0.50	1,612	Litres	IS 456 Cl 5.4
Surface Primer	Water-thinnable primer @ 8 m²/L (1 coat)	106.1	Litres	NBC 2016 SP7
Polymer Wall Putty	2 coats (1–2 mm) @ 1.5 kg/m²	1,533.2	kg	NBC 2016 SP7
Interior Acrylic Emulsion	2 coats (30–40 microns) @ 12 m²/L	105.2	Litres	NBC 2016 SP7
Exterior Weather Emulsion	2 coats (40–60 microns) @ 10 m²/L	43.6	Litres	NBC 2016 SP7
GRAND TOTAL CEMENT (Mortar + Plaster)	All masonry + plastering works combined	166 bags (8,251 kg)	Bags	IS 269:2015
GRAND TOTAL SAND (Mortar + Plaster)	All masonry + plastering sand combined	38.76	Tonnes	IS 383:2016
GRAND TOTAL WATER (Mortar + Plaster)	All masonry + plastering water combined	4,125	Litres	IS 456 Cl 5.4

11.2 Trade-wise Cost Takeoff

Trade / Material Component	Quantity	Rate (INR)	Subtotal (INR)
Clay Modular Bricks (IS 1077)	17,512 Nos	■9 / brick	■157,608.00
Brickwork Mortar Cement	101 Bags	■380 / bag	■38,380.00
Brickwork Mortar Sand	22.34 Tonnes	■1,500 / Tonne	■33,506.45
Plastering Work (12/15/6mm)	848.9 m²	■280 / m²	■237,678.00
Surface Finishes (Primer + Putty + Paint)	848.9 m²	■120 / m²	■101,862.00
TOTAL MASONRY, PLASTER & FINISHES			■569,034.45

12. Structural Layout Plan

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Column Coordinates: C0_L0: (0.00, 0.00) C1_L0: (3.30, 0.00) C2_L0: (6.60, 0.00) C3_L0: (9.90, 0.00)
C4_L0: (0.00, 4.00) C5_L0: (3.30, 4.00) C6_L0: (6.60, 4.00) C7_L0: (9.90, 4.00) C8_L0: (0.00, 8.00)
C9_L0: (3.30, 8.00) C10_L0: (6.60, 8.00) C11_L0: (9.90, 8.00)
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13. Integrated Steel Breakdown

Bar Diameter	Total Weight (kg)
10 mm	0.0
Est_Beam	0.0
Est_Col	505.4

Bar Diameter	Total Weight (kg)
TOTAL	505.4

14. Structural Audit Report

No audit results available.

15. Sustainability Audit

Design Strategy: Standard (OPC)

Material	Quantity	Emission Factor	Total CO2e (kg)
Concrete	3.24 m3	300 kg/m3	972.0
Steel	505 kg	1.85 kg/kg	935.1
TOTAL	-	-	1907.1

Wait! You can reduce your Carbon Footprint.
Refining the design to use **Fly Ash based Concrete (PPC)** instead of OPC can reduce concrete emissions by approximately 15-20%.
Switch "Use Green Concrete" ON in the dashboard to see savings. (Note: PPC is slower setting and has lower early strength, use only if confirmed by RMC supplier).

16. Serviceability Checks (Deflection)

No beam serviceability checks performed.

17. Staircase Design Schedule (Dog-Legged)

No staircase selected for design.

20. Concrete Material Breakdown (IS 10262:2019)

Material	Per m³	Total
Wet Concrete Volume		3.24 m³
Cement (50kg bags)	7.6 bags	24.6 bags
Fine Aggregate (Sand)	0.419 m³	1.36 m³
Coarse Aggregate (20mm)	0.821 m³	2.66 m³
Water	171 kg	554 kg
Water:Cement Ratio	0.45	—
Nominal Mix Ratio	1:1.76:3.13	—

Note: Indicative quantities per IS 10262:2019. Add 3-5% wastage for actual procurement.

24. Professional Disclaimer

IMPORTANT NOTICE - READ CAREFULLY

This structural design report is generated by automated software for **PRELIMINARY DESIGN AND EDUCATIONAL PURPOSES ONLY**.

Key Limitations:
• The software uses simplified analysis methods (tributary area, short column assumptions).
• Seismic forces (IS 1893) and Wind loads (IS 875 Pt 3) have been incorporated for basic checks, but require full dynamic analysis for high-rise buildings.
• Site-specific soil conditions, groundwater, and other geotechnical factors are not evaluated.
• Local building codes may have additional requirements beyond IS 456:2000.

Required Actions Before Construction:

- All designs MUST be reviewed and sealed by a licensed Professional Engineer (PE).
- Independent structural analysis should be performed using appropriate software.
- Geotechnical investigation and foundation recommendations are required.
- All applicable permits and approvals must be obtained.

Liability Disclaimer:

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Generated by StructOptima. This report is a preliminary design aid compliant with IS 456:2000, IS 1893:2016, IS 13920:2016.
All designs must be verified by a licensed Structural Engineer prior to construction.
Safety Factors: γc = 1.5 (concrete), γs = 1.15 (steel) | Load Factor: 1.5 (DL + LL)
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